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Literature & Technology Review

TARAS: Agricultural Analysis & Simulation

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TABLE OF CONTENTS

1. Introduction....3
    1.1. Background....3
    1.2. Problem Definition....3
    1.3. Scope of the Study....5
    1.4. Aim and Contributions....5
2. IoT Sensor Node Architecture....6
    2.1. Microcontroller Selection: Fully Wireless ESP32....6
    2.2. Sensing Technologies: Capacitive vs. Resistive....7
    2.3. Sensor Calibration....7
3. Cloud Computing and Data Management....7
    3.1. Structured Database Design (PostgreSQL)....8
4. Irrigation Recommendation....8
5. Artificial Intelligence and Machine Learning....9
    5.1. Computer Vision for Disease Detection (CNNs)....9
    5.2. Large Language Models....9
6. Visualization and Digital Twins....10
    6.1. Reducing Cognitive Load with Digital Twins....10
7. Mobile Application....10
    7.1. Mobile Applications in Agriculture....10
    7.2. Mobile App in TARAS....10
    7.3. Technology Stack....11
8. Ecological Impact....11
    8.1. Water Consumption....11
    8.2. Carbon Footprint....12
9. Related Work....13
    9.1. Single-Purpose Smart Irrigation Systems....13
    9.2. Standalone Disease Detection Applications....13
    9.3. High-End Digital Twins....14
10. Conclusion and Research Gap....14
11. References....14

1. Introduction

Modern agriculture is increasingly affected by climate variability, declining freshwater resources, and sustainability challenges. Many farmers still make irrigation and crop-management decisions based on intuition rather than systematic measurements, which reduces decision accuracy and leads to inefficient water use [1]. Research also shows that improper irrigation practices can cause water loss and yield reduction, making data-driven decision support essential for future agricultural productivity [2], [3].

Recent advances in digital agriculture offer viable solutions. IoT-based sensing systems provide continuous measurements of soil and environmental conditions [4], while machine-learning algorithms can analyze these data to generate accurate irrigation recommendations [4]. Digital twin technologies enable virtual representations of agricultural fields for real-time simulation and visualization [5], and later studies show that digital twins can significantly enhance field monitoring under dynamic conditions [6]. In parallel, computer-vision models have demonstrated strong performance in diagnosing plant diseases directly from leaf images [7].

TARAS platform brings these technologies together into a unified, farmer-oriented system that offers real-time field monitoring, irrigation prediction, disease assessment, and digital-twin visualization. The project aligns with Türkiye's Twelfth Development Plan, which emphasizes AI-supported digital agriculture [8], and also contributes to global sustainability objectives such as SDG 12, SDG 13, and SDG 15 [9].

1.1. Background

Agriculture has progressed through four major transformations: the emergence of settled farming, the mechanization introduced during the Industrial Revolution, the productivity gains of the Green Revolution, and today's Agriculture 4.0 era driven by digital technologies. Precision Agriculture, a key component of this era, relies on monitoring and responding to spatial and temporal field variability to optimize resource use [4]. IoT sensors enable real-time environmental monitoring, digital twins provide interactive virtual models of fields for prediction and analysis [5], and computer-vision systems support early disease recognition through image-based diagnosis [7]. TARAS builds on this technological landscape to create a unified platform tailored for real agricultural environments.

1.2. Problem Definition

Traditional Farming Limitations: One of the major challenges in modern agriculture is that farmers struggle to make reliable, data-based decisions. Many still rely on intuition rather than systematic and evidence-based methods, which limits the accuracy of field management [1]. Although intuition has supported

farming for generations, it is becoming less effective as climate patterns shift and soil conditions change. As a result, crops often fail to reach their full yield potential when decisions depend solely on historical experience.

Resource Scarcity: Water scarcity is one of the most critical constraints in agriculture today. Traditional irrigation methods frequently lead to over-watering, which wastes water and damages root systems, or under-watering, which stunts plant growth. Studies show that inefficient or poorly timed irrigation can cause significant water loss and reduce productivity [2], [3]. Because soil moisture varies across fields and changes over time, static rule-based irrigation schedules cannot meet crop-specific requirements, leading to substantial inefficiencies in both water and energy use.

Disease Management Challenges: Farmers commonly notice unusual visual changes in plants but cannot determine whether these differences indicate disease or harmless variation. This uncertainty can delay intervention or lead to unnecessary pesticide use. When early symptoms are subtle or unfamiliar, problems may be ignored until they become severe, or pesticides may be applied without need. Both outcomes increase production costs and negatively affect food safety and environmental health. Image-based decision-support systems can help address this uncertainty by allowing farmers to upload leaf images and receive automated analysis, improving early assessment and supporting timely, informed action [7].

Data Quality & System Reliability Issues: Even when farmers adopt technological tools, field data are often incomplete or inconsistent due to sensor drift, hardware faults, or communication failures. Soil-moisture probes may lose calibration over time, wireless connections may drop, and environmental fluctuations can distort sensor readings. Without proper validation, calibration, and anomaly-detection mechanisms, inaccurate data can mislead decision-making and become as harmful as intuition-based irrigation.

Lack of Accessible, Integrated Decision Support: Many farms rely on fragmented digital tools, such as separate systems for sensors, irrigation controllers, weather forecasts, and crop health information. Because these systems do not communicate with each other, farmers, especially small-scale producers, struggle to benefit from advanced analytics. This fragmentation highlights the need for an integrated platform that combines real-time field monitoring, intelligent recommendations, visual explanations, and sustainability metrics within a single accessible system.

1.3. Scope of the Study

The scope of TARAS covers the development of an end-to-end precision-agriculture platform. The system includes IoT-based soil-moisture and temperature sensing using ESP32 microcontrollers with integrated battery management, and direct WiFi connectivity for field-to-cloud data transfer. It incorporates a secure cloud database for storing real-time and historical records, physics-based irrigation prediction algorithms following FAO guidelines, and computer-vision algorithms for disease identification.

The project further includes the design of a mobile application that provides farmers with moisture visualization through a digital twin using Mapbox-based SVG overlays, image-based disease assessment with weather-integrated risk prediction, and an AI-assisted decision interface. The system focuses on decision support rather than full automation, ensuring that farmers remain central to the decision-making process. It is intended to be low-cost, portable, and scalable across various field configurations.

1.4. Aim and Contributions

The aim of TARAS is to provide farmers with an intelligent, real-time agricultural assistant that can analyze environmental conditions, predict irrigation requirements, detect plant diseases from leaf images, and visualize moisture distribution through digital twin technology. By bringing together IoT, machine learning, computer vision and cloud computing, the system offers a highly adaptive and user-centered approach that supports precision farming and sustainable resource management.

TARAS aims to optimize irrigation efficiency, reduce water and energy waste, enhance crop health monitoring, improve sustainability through carbon footprint tracking and provide farmers with actionable, explainable, and accessible insights

Contributions: Recent advancements in smart agriculture highlight the value of combining IoT-based sensing with machine-learning-driven decision systems to improve irrigation efficiency and field management. Studies show that ML algorithms using soil moisture, temperature, humidity, and soil-type data can generate highly precise irrigation recommendations and adapt to changing field and weather conditions. TARAS builds on these findings by pairing such predictive models with real-time digital-twin visualizations, enabling farmers to interpret moisture patterns through intuitive heatmaps, observe changes over time, and identify areas requiring intervention more clearly.

Beyond irrigation, TARAS incorporates computer-vision-based disease detection that allows farmers to upload leaf images through the mobile

application. These images are processed using CNN models that classify diseases or stress symptoms and provide actionable assessments, reducing the need for expert diagnosis. The platform also addresses sustainability by integrating carbon-footprint metrics, estimating emissions based on pump energy usage, fertilizer inputs and water savings.

By combining these capabilities into a single, cohesive platform, TARAS contributes to the development of intelligent, adaptive and sustainability-focused digital farming systems that make advanced decision support accessible to farmers across different scales.

2. IoT Sensor Node Architecture

The foundation of any smart farming system is the ability to collect accurate environmental data from the field [10]. The literature emphasizes that hardware in agriculture must be low-power, durable, and cost-effective.

2.1. Microcontroller Selection: Fully Wireless ESP32

In the realm of agricultural IoT, the primary engineering challenge is maximizing the operational lifespan of battery-powered nodes. Zhang and Meng (2021) highlight that energy efficiency is the most critical factor for the economic viability of precision agriculture, as frequent battery replacements increase labor costs significantly [11], and using wired technology introduces complexity most fields are not suited to deal with.

While earlier projects successfully utilized standard ESP32 or ESP8266 modules, recent literature indicates a shift toward more specialized architectures. Recent studies in wireless sensor networks have begun favoring efficient architectures for their superior power-to-performance ratio. For example, a study by Irga et al. (2023) implemented agricultural monitoring nodes noting a significant reduction in standby power consumption compared to traditional controllers [12].

Building on these findings, the TARAS project utilizes the ESP32 microcontroller, which offers excellent power efficiency, and by integrating a battery, allows for a fully wireless solution. This integrated approach simplifies the hardware design while ensuring reliable long-term operation in field conditions. Mohd Khalid et al. (2024) highlighted that these chips, when paired with renewable energy sources, allow nodes to survive extended periods without going offline, a critical factor for continuous year-round field monitoring [13].

Another advantage of using ESP32 is its support of the I²C (Inter-Integrated Circuit) Communication Interface, which is a widely used two-wire communication interface designed for low-power embedded systems. I²C enables multiple digital sensors to communicate with the ESP32 microcontroller over a shared SDA (data) and SCL (clock) bus, reducing wiring complexity and supporting scalable node designs. Its address-based structure allows several sensors to operate simultaneously, making it particularly suitable for multi-parameter agricultural monitoring. The protocol's low latency and efficient power consumption make it ideal for continuous field measurements where reliability and simplicity are critical.[14]

2.2. Sensing Technologies: Capacitive vs. Resistive

Accurate irrigation depends entirely on accurate soil moisture readings. Historic literature often cited "resistive" sensors, which measure moisture by passing an electric current through the soil [15]. However, engineering studies have extensively documented the failure of these sensors due to **electrolysis**, which causes the metal probes to corrode (rust) within weeks [16].

To solve this, modern research advocates for **Capacitive Soil Moisture Sensors**. These measure the dielectric permittivity of the soil, basically how much charge the soil can hold, without any exposed metal touching the soil. This prevents corrosion and ensures reliable data over long periods [17]. Similarly, for air metrics, digital sensors like the **SHT Series sensors** are preferred over analog thermistors because they provide calibrated digital output, eliminating errors caused by electrical noise in the wires [18].

2.3. Sensor Calibration

Proper calibration of soil moisture sensors is essential for accurate readings. The TARAS system employs a simple min-max calibration procedure that can be performed in the field. Using two reference samples, completely dry soil and fully saturated (muddy) soil, the sensor's raw analog values are mapped to the 0-100% moisture range. This calibration data is stored and applied to all subsequent readings. This approach ensures that sensors remain accurate across different soil types and conditions without requiring complex laboratory calibration procedures.

3. Cloud Computing and Data Management

Handling the influx of data from the field requires a scalable and organized backend system.

3.1. Structured Database Design (PostgreSQL)

Agricultural data is primarily "Time-Series Data" (e.g., specific moisture values recorded at specific times). While flexible NoSQL databases are popular for web apps, scientific applications usually prefer Relational Databases like **PostgreSQL** for their strict structure and reliability. PostgreSQL is ACID-compliant, meaning it guarantees data validity even in the event of errors or power failures [24]. Furthermore, the system utilizes a "Log-Based" architecture. As recommended in foundational database literature, this approach involves never deleting or overwriting old records, but simply appending new ones [24]. This creates an unchangeable history of the field, which is essential for training accurate machine learning models later.

4. Irrigation Recommendation

Determining when to irrigate crops is primarily a decision-making problem that depends on accurately interpreting soil-moisture dynamics and environmental conditions. Traditional irrigation controllers often rely on fixed schedules or simple threshold-based rules, which can lead to inefficient water use when field conditions vary over time.

In this system, irrigation decisions are generated using a calibrated rule-based approach that evaluates time-series soil-moisture trends together with environmental conditions and field-specific parameters. By analyzing moisture-level thresholds and rate-of-change behavior, the method adapts irrigation timing and duration without relying on large historical datasets or complex predictive models.

Compared to machine-learning-based solutions, this rule-based method offers improved transparency, lower computational complexity and greater robustness in environments with limited data availability or intermittent connectivity. Its explainable structure allows irrigation recommendations to be easily interpreted and adjusted, making it suitable for practical field deployment.

5. Artificial Intelligence and Machine Learning

Collecting data is only useful if it leads to better decisions. **Artificial Intelligence (AI)** is used to process raw numbers into actionable advice [25].

5.1. Computer Vision for Disease Detection (CNNs)

Identifying plant diseases early can prevent significant yield losses; however, expert agronomists are not always readily available [7]. To address this challenge, we employ Convolutional Neural Networks (CNNs), which are deep learning models specifically designed for image analysis. By training a CNN on large-scale datasets containing images of both healthy and diseased plant leaves, the model learns to identify discriminative visual patterns such as color variations, texture irregularities, and lesion structures associated with different plant diseases.

In this study, the **EfficientNet architecture** is selected due to its ability to achieve high classification accuracy with significantly fewer parameters compared to conventional convolutional neural network (CNN) models. EfficientNet introduces a compound scaling strategy that uniformly scales network depth, width, and input resolution, leading to improved performance while preserving computational efficiency. This balanced scaling approach has been shown to outperform traditional CNN architectures across various image classification benchmarks while requiring fewer computational resources.[19]

5.2. Large Language Models

To translate complex analytical outputs into actionable guidance for farmers, TARAS integrates a **Large Language Model (LLM)** within an interactive chatbot interface. This component acts as the system's communication layer, interpreting sensor data, historical irrigation records, and computer vision findings to generate clear, context-aware explanations.

Research by Wang et al. (2024) indicates that LLMs significantly bridge the gap between technical data and farmer understanding by providing natural language explanations for agronomic advice [20]. Instead of presenting users with raw numerical values, the LLM-equipped assistant provides meaningful insights, for example, why irrigation is recommended on a specific day or how a detected disease might impact the crop. Additionally, the LLM supports "what-if" scenario analysis by simulating alternative decisions (such as delaying irrigation or applying preventive treatments) and explaining their potential outcomes. Through domain-adapted fine-tuning or retrieval-augmented generation (RAG), the model

ensures high accuracy, transforming TARAS into a transparent, reliable, and user-friendly agricultural advisor.

6. Visualization and Digital Twins

The interface between the human and the machine is just as important as the technology itself.

6.1. Reducing Cognitive Load with Digital Twins

"Cognitive Load" refers to how much mental effort is needed to understand information. Research indicates that standard spreadsheets impose a high cognitive load, whereas visual maps reduce this load significantly [21]. The concept of the **Digital Twin** involves creating a 3D virtual replica of the physical field. Research suggests that visualizing data spatially, such as a 3D heat map where dry areas glow red, allows farmers to identify problem zones much faster than reading lists of numbers [22]. To make this accessible, **React Native** is used to build a cross-platform mobile app, ensuring the tool works on the smartphones farmers already own.

7. Mobile Application

Mobile applications have emerged as one of the most widely adopted interfaces in smart agriculture systems. Studies highlight that mobile platforms make complex IoT and AI infrastructures accessible by presenting field information directly to farmers in real time. They support continuous monitoring, rapid decision making, and convenient data entry, which makes them a practical tool for field operations where desktop interfaces are rarely available [23].

7.1. Mobile Applications in Agriculture

Recent smart-agriculture research emphasizes the role of mobile applications as the primary interaction layer between farmers and digital farming systems. Mobile tools frequently act as gateways for sensor networks, cloud platforms, and machine-learning models. They are used for monitoring field conditions, visualizing environmental data, delivering agronomic recommendations, and providing remote diagnostic services. Literature also notes that mobile interfaces significantly increase technology adoption among farmers, especially when the design is simple and the recommendations are easy to interpret [23].

7.2. Mobile App in TARAS

Within this literature landscape, the TARAS mobile application follows the same principles identified in prior studies. It functions as the primary interface through which farmers receive sensor readings, irrigation recommendations, and disease analysis results. The app also includes a 3D digital-twin visualization that represents moisture distribution spatially, a feature that extends the capabilities of many existing mobile agricultural systems. By combining monitoring, prediction, disease assessment, and spatial visualization into a single platform, TARAS addresses the fragmentation noted in current literature and provides a unified decision-support tool aligned with modern precision-agriculture practices.

7.3. Technology Stack

The technological choices used in the TARAS mobile application reflect practices recommended in mobile and IoT literature. **React Native** is widely adopted for agricultural applications because it enables cross-platform deployment on Android and iOS, which is essential given the diversity of devices in rural areas [24]. Expo simplifies development and maintenance processes by managing native modules and supporting incremental updates, which is useful for systems deployed in field environments with limited technical infrastructure [24].

For visualization, **React Three Fiber** allows efficient rendering of 3D elements on mobile devices, supporting interactive digital-twin displays similar to those explored in recent spatial agriculture studies [22]. The data exchange between the mobile client and the cloud backend uses a simple request–response API model, which is frequently adopted in smart farming architectures for real-time synchronization.

8. Ecological Impact

Technology must also serve environmental goals. Modern agriculture is under pressure to reduce its environmental impact, in line with UN Sustainable Development Goals 12 and 13 [9]. Literature suggests that IoT systems are uniquely positioned to automate sustainability tracking. By monitoring exactly how long water pumps run and how much resource is consumed, the system can calculate a real-time **Carbon Footprint**. This moves sustainability from a vague concept to a measurable metric [25].

8.1. Water Consumption

The TARAS system promotes responsible resource use by ensuring that irrigation is conducted only when and where it is needed, directly supporting the principles of Goal 12. By replacing uniform, schedule-based watering with sensor-driven and Algorithm-optimized recommendations, TARAS minimizes unnecessary water consumption, reduces nutrient runoff, and lowers the overall

environmental footprint of agricultural production. This precision in irrigation not only conserves water but also decreases the energy required for pumping, thereby reducing operational emissions and contributing to Goal 13 on climate action. Through its Digital Twin visualizations and adaptive decision-support engine, the system enables farmers to monitor consumption patterns, identify inefficiencies, and implement more sustainable long-term practices. In doing so, TARAS strengthens both resource responsibility and climate resilience within modern agriculture.

8.2. Carbon Footprint

Agriculture is one of the most significant contributors to global greenhouse gas (GHG) emissions, accounting for approximately one-fifth of total anthropogenic emissions [26]. The main sources include fertilizer production and application, energy consumption for irrigation and machinery, livestock management, and soil-based emissions such as nitrous oxide (N_2O) from nitrogen fertilizers and methane (CH_4) from organic decomposition. These emissions, when expressed as carbon dioxide equivalents (CO_2e), constitute the carbon footprint of agricultural production. Understanding and reducing this footprint has become a major focus in achieving the United Nations Sustainable Development Goals (SDGs), particularly Goal 12 (Responsible Consumption and Production) and Goal 13 (Climate Action).

Recent studies indicate that digital and IoT-based farming systems can play a transformative role in mitigating agricultural greenhouse gas emissions. Precision irrigation systems that utilize real-time soil-moisture and climate sensing enable farmers to reduce unnecessary water use and the associated energy required for pumping, thereby decreasing CO_2 emissions [27]. Similarly, IoT-supported fertilization and nutrient-management approaches help align nitrogen application with actual crop demand, preventing overuse and limiting nitrous oxide (N_2O) emissions, one of the most potent agricultural greenhouse gases [28]. Overall, these digital and sensor-driven strategies improve resource efficiency and contribute to more sustainable agricultural production [29].

ML and decision-support systems further enhance the ability to estimate and monitor carbon footprints dynamically. By combining environmental sensor data (soil moisture, temperature) with user-provided activity data (fertilizer use, diesel consumption, electricity usage), these systems can calculate real-time carbon emission scores at the farm or greenhouse level.

Integrating this functionality into a mobile application significantly increases accessibility and user engagement. Farmers can view their estimated emissions, identify the main contributing factors. Within the TARAS system, this allows users to enter their input data and instantly calculate their farm's carbon

footprint. The module supports sustainability awareness and provides actionable insights for emission reduction strategies, reinforcing the system's contribution to climate action (Goal 13) and responsible production (Goal 12).

9. Related Work

To design the TARAS platform, we analyzed several existing systems and research projects. While many solutions exist, they typically solve only one part of the problem. This section discusses these related works and how they directly influenced our design choices.

9.1. Single-Purpose Smart Irrigation Systems

A significant amount of research focuses solely on automating water pumps.

- **The Study:** A prominent study by Nawandar & Satpute (2019) developed a low-cost IoT system using simple neural networks to schedule irrigation based on soil data [30].
- **Limitation:** While effective at saving water, this system operates in a vacuum. It assumes the plant is healthy. If a plant is attacked by a fungus, it might need less water or a chemical treatment, not just standard irrigation. The system had no way of "seeing" the plant's health.
- **Effect on TARAS:** This influenced us to build a hybrid system. TARAS combines soil sensors with camera-based disease detection so decisions are made based on the whole picture, not just the soil moisture level.

9.2. Standalone Disease Detection Applications

On the other side of the spectrum, there is extensive research on using AI to spot diseases.

- **The Study:** The most famous work in this field is by Mohanty et al. (2016), who trained Deep Learning models on the PlantVillage dataset [7].
- **Limitation:** While scientifically impressive, this approach was purely diagnostic. It tells the farmer "You have Apple Scab," but it doesn't link that information to the farm's environmental history. Was the disease caused by over-watering? The app couldn't say because it didn't have access to soil data.
- **Effect on TARAS:** This taught us that context matters. In TARAS, when a disease is detected, the system can cross-reference it with historical humidity logs to help explain why the outbreak happened.

9.3. High-End Digital Twins

Finally, we looked at existing "Digital Twin" concepts.

- **The Study:** Pylianidis et al. (2021) explored creating digital twins for high-tech greenhouses. These systems create incredible 3D simulations of crop growth and climate control [31].
- **Limitation:** These systems are extremely complex, expensive, and require powerful desktop computers. They are designed for industrial agronomists, not for a typical farmer standing in a field with a phone.
- **Effect on TARAS:** This defined our Mobile-First strategy. We chose technologies like React Native and low-poly 3D rendering to ensure our Digital Twin works smoothly on a standard smartphone, making the technology accessible to regular farmers.

10. Conclusion and Research Gap

However, a significant gap remains. Most current solutions function in isolation, offering either irrigation control or disease detection, and often lack user-friendly visualization tools for small-scale farmers. Furthermore, few systems actively track sustainability metrics like carbon footprints alongside agronomic data. Consequently, there is a need for a unified, low-cost platform that combines real-time sensing, AI-driven predictions and visual analytics into a single accessible mobile system. The proposed TARAS project aims to address this gap by developing an end-to-end solution tailored for these needs.

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